

1(a)	stem / not a root, because, transport tissue / vascular bundles / vascular tissue(s) / xylem and phloem / xylem / phloem, around the edges / not in the centre / AW ; A stem because, pith / parenchyma tissue, is in the centre	1
1(b)	<i>any two from:</i> (hydrostatic) support / described ; R if incorrect context e.g. support because of thick cell walls A packing tissue (cells), turgid / store <u>water</u> ; storage ; <i>in context of substances other than water e.g. sucrose / starch / waste</i>	2
1(c)(i)	vascular bundle ;	1
1(c)(ii)	0.65 (<u>mm</u>) ;	1

Question	Answer	Marks																
1(d)	<i>mp1-3 must have statements for both xylem and phloem</i> <i>mp4-7 need statements for either xylem or phloem</i> <i>three from:</i> <table><thead><tr><th><i>xylem</i></th><th><i>phloem</i></th></tr></thead><tbody><tr><td>1 no cytoplasm / hollow / no contents</td><td>(peripheral / little / some) cytoplasm ; I protoplasm R full of cytoplasm / AW</td></tr><tr><td>2 no organelles / hollow / no contents</td><td>(a few) organelles ; A examples of organelles mitochondria or ER I has SER / RER / ribosomes R has nucleus</td></tr><tr><td>3 lignified R idea of lignin within element</td><td>no lignin / (only) cellulose ;</td></tr><tr><td>4 no end wall(s) / no sieve plate(s) A end walls have broken down</td><td>or sieve plate(s) / perforated end wall(s) ; A (end walls) have sieve pores</td></tr><tr><td>5 (side walls) contain pits</td><td>or no pits (in side walls) ;</td></tr><tr><td>6 no plasmodesmata</td><td>or plasmodesmata (to companion cells) ;</td></tr><tr><td>7 thick(er), cell wall / walled</td><td>or thin(ner), cell wall / walled ;</td></tr></tbody></table>	<i>xylem</i>	<i>phloem</i>	1 no cytoplasm / hollow / no contents	(peripheral / little / some) cytoplasm ; I protoplasm R full of cytoplasm / AW	2 no organelles / hollow / no contents	(a few) organelles ; A examples of organelles mitochondria or ER I has SER / RER / ribosomes R has nucleus	3 lignified R idea of lignin within element	no lignin / (only) cellulose ;	4 no end wall(s) / no sieve plate(s) A end walls have broken down	or sieve plate(s) / perforated end wall(s) ; A (end walls) have sieve pores	5 (side walls) contain pits	or no pits (in side walls) ;	6 no plasmodesmata	or plasmodesmata (to companion cells) ;	7 thick(er), cell wall / walled	or thin(ner), cell wall / walled ;	3
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